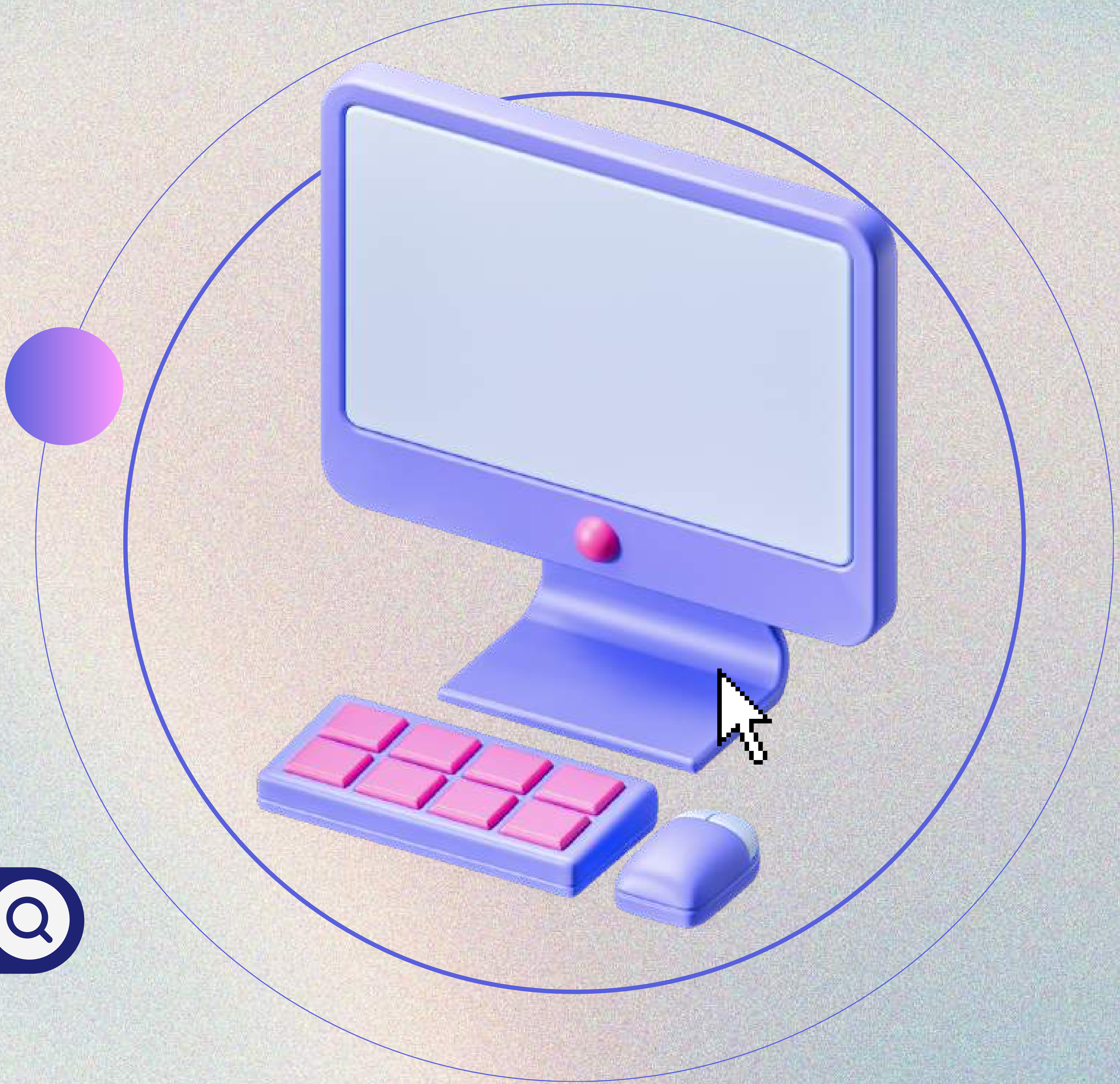
AWD - CodeSync

DATA MINING & BIG DATA



Team: Danna Andrade, Danny Ayuquina, Xavier Altamirano

BIG DATA

It focuses on the volume, velocity, variety, and accuracy of information. Its goal is to organize large amounts of data that traditional tools cannot handle.

A technical, automatic or semi-automatic process that analyzes large amounts of scattered information to give it meaning and turn it into knowledge.

7 V OF BIG DATA

Volume,

Velocity,

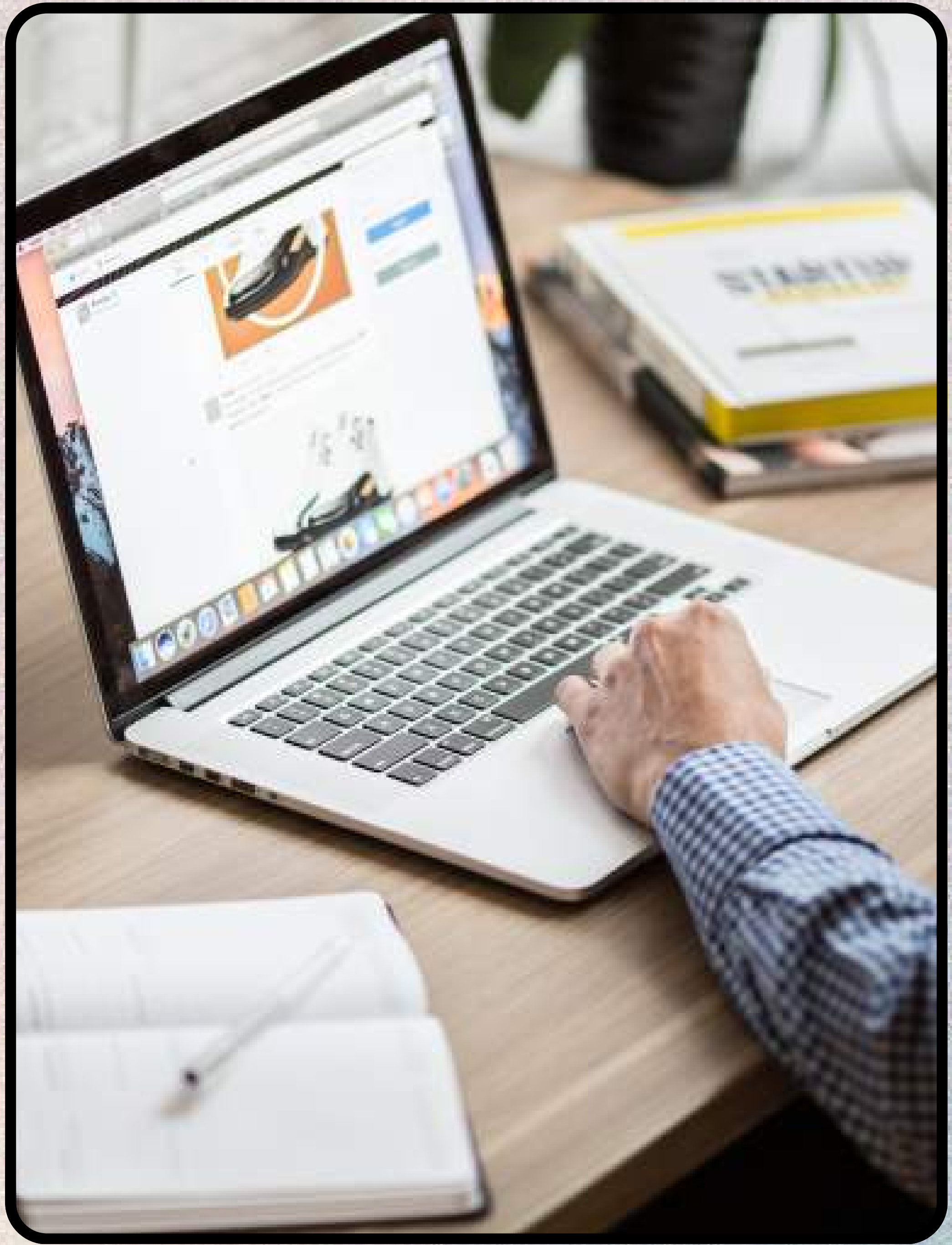
Variety,

Veracity,

Value,

Visualization, and

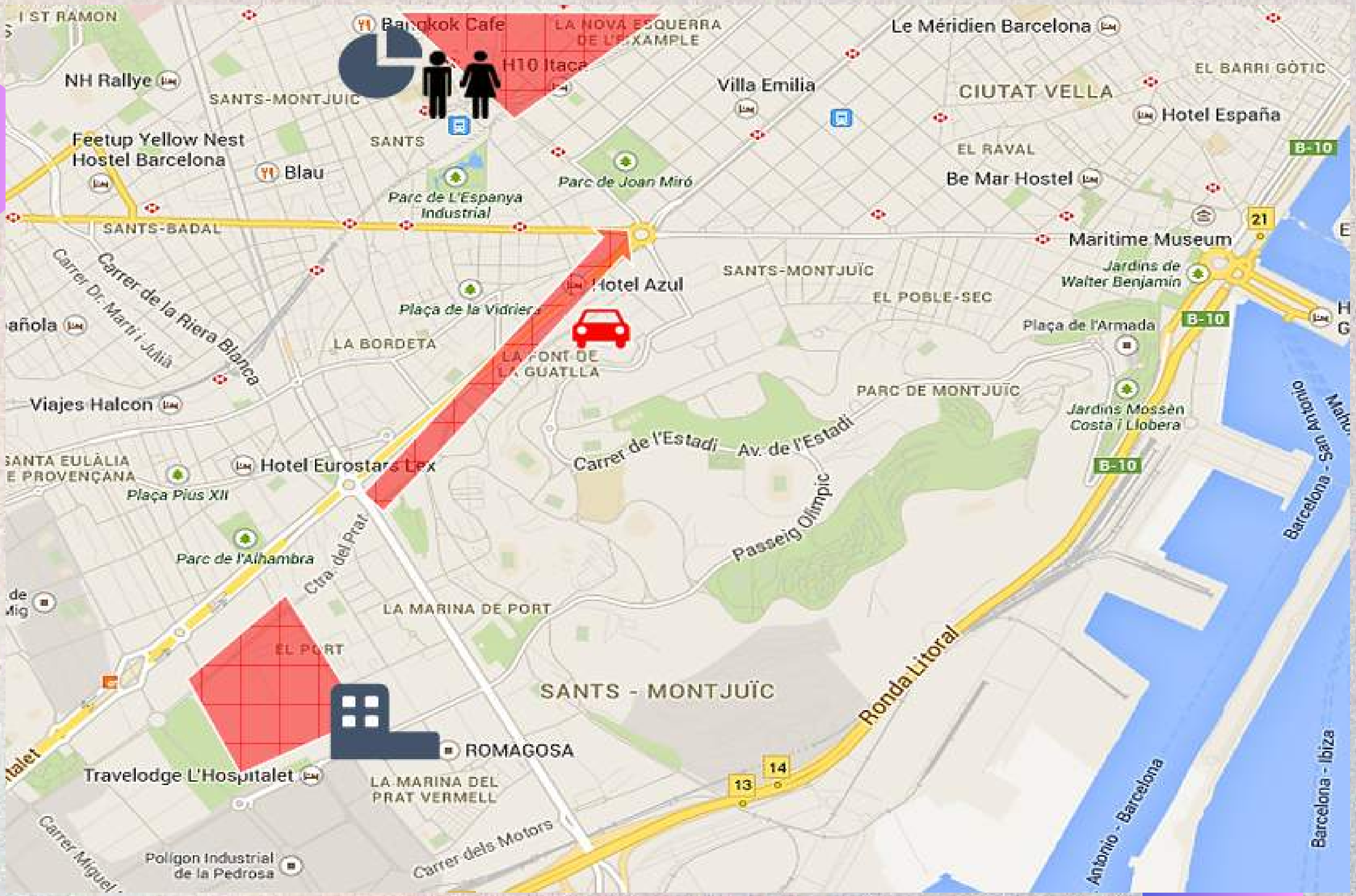
Viability/Variability



MAPS

Footprints
People movement patterns and dwell patterns • Hourly and daily analysis • Inference for dwell metrics within buildings

Sociodemographics and Behavior
Sociodemographic profile combined with movement ,Age, gender, country, home address, work address, type of device (data, status), activity patters



PROPOSAL

OBJECTIVE

Implement an analytics platform to collect, store, and analyze data generated by the daycare to support operational and strategic management.

The Big Data proposal focuses on:

- Capturing data from multiple daycare processes
- Storing historical and structured information
- Monitoring system behavior over time
- Generating key performance indicators (KPIs) for decision-making

PROPOSAL

KPI’S (KEY PERFORMANCE INDICATOR)

- **DAILY AND WEEKLY ATTENDANCE RATE**
- **PUNCTUALITY BY CLASSROOM**
- **OCCUPANCY AND AVAILABLE CAPACITY**
- **REVENUE VS. DELINQUENCY RATE**
- **AVERAGE PAYMENT TIME**
- **COMMUNICATION EFFECTIVENESS**
- **INCIDENTS AND RESOLUTION TIME**

PROPOSAL

DATA TO CAPTURE

- Attendance: entries, exits, punctuality, absences
- Classrooms and capacity: spaces, occupancy, peak times
- Finance: payments, late payments, billing, discounts
- Communication: messages sent, opened, and replied to
- Incidents: health, behavior, follow-ups
- System usage: events, errors, loading times

ARCHITECTURE

- Ingestion: Backend and web events
- Storage:
- Data Lake (historical)
- Data Warehouse (analytical queries)
- Processing: Scheduled ETL/ELT
- Visualization: Dashboards and alerts (BI)

PROPOSAL

BENEFITS

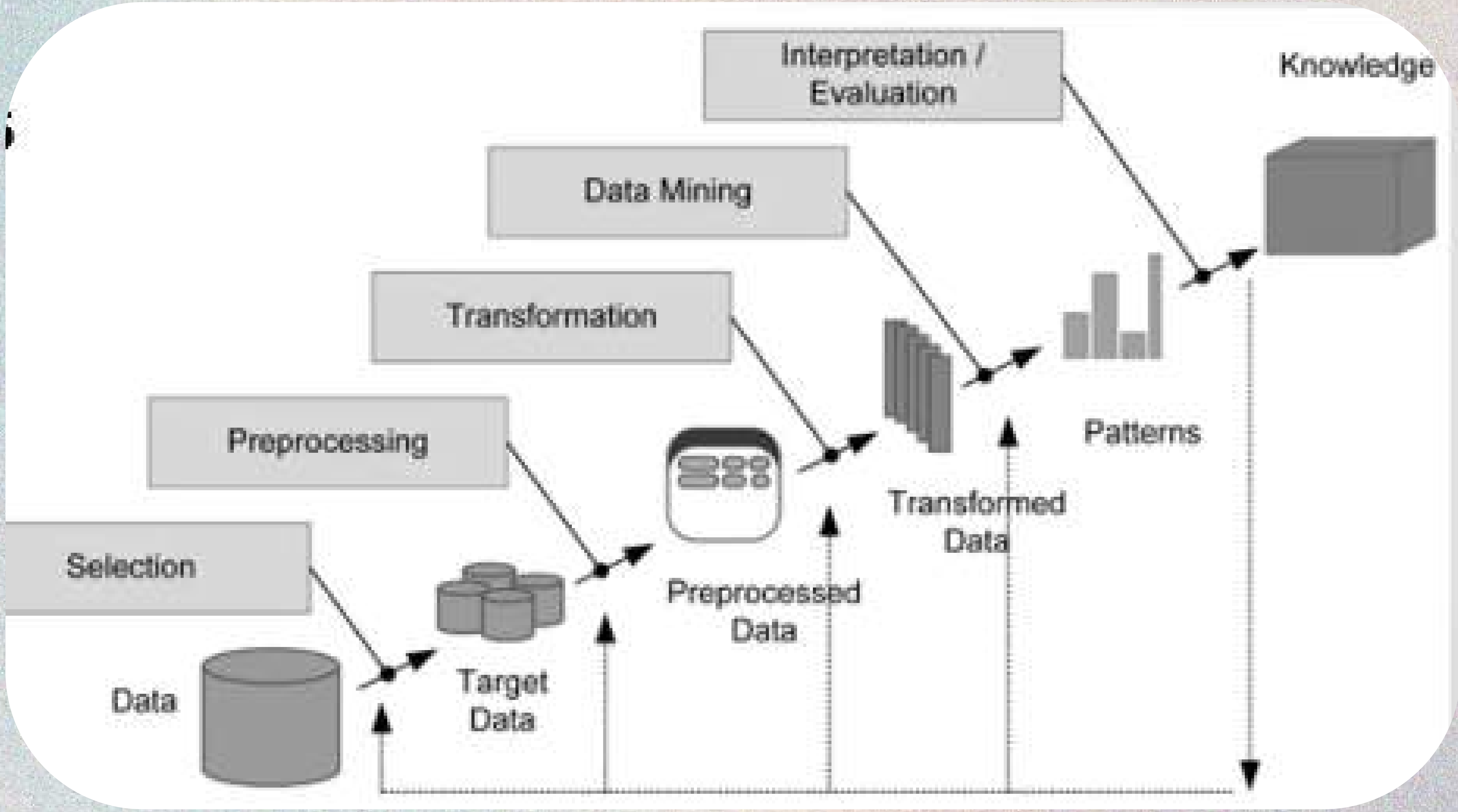
With Big Data, the daycare can:

- Analyze attendance, punctuality, and absenteeism
- Evaluate classroom occupancy and capacity
- Monitor payments and late payments
- Measure communication effectiveness with families

WHAT IS DATA MINING?

Data mining is the process of extracting useful patterns, trends, or insights from large sets of structured or unstructured data. It involves various techniques, such as statistical analysis, machine learning, and artificial intelligence, to identify meaningful patterns or relationships within the data.

DATA MINING PROCESS



Data Mining process, also known as the KDD process, which means Knowledge Discovery in Databases.

EXAMPLE

DATA MINING IN E-COMMERCE

E-commerce leaders like Amazon mine past purchase data, browsing behavior, and customer similarity scores to drive real-time product suggestions, a tactic responsible for up to 35% of Amazon's revenue.

"Customers who bought this also liked..." is powered by association rules, clustering, and collaborative filtering, all data mining techniques.

PROPOSAL

OBJECTIVE

Apply advanced analytics techniques to discover patterns and make predictions that improve planning and decision-making.

The Data Mining proposal enables:

- Identifying behavioral patterns
- Predicting absenteeism risk
- Segmenting families based on engagement
- Detecting operational anomalies

PROPOSAL

DATA TO RECOLECT

- Child attendance history
- Age, classroom, school calendar
- Payments and outstanding balances
- Communications interaction
- Incidents and observations
- External variables (weather, events)

METHODOLOGY

1. Data cleaning and preparation
2. Feature creation
3. Model training
4. Validation with metrics
5. Results interpretation

PROPOSAL

BENEFITS

Analytical models help to:

- Anticipate issues before they occur
- Optimize staff allocation
- Personalize communication with families
- Provide timely support to children and parents

THANK
YOU!